NomNom

The Smart ChefBot

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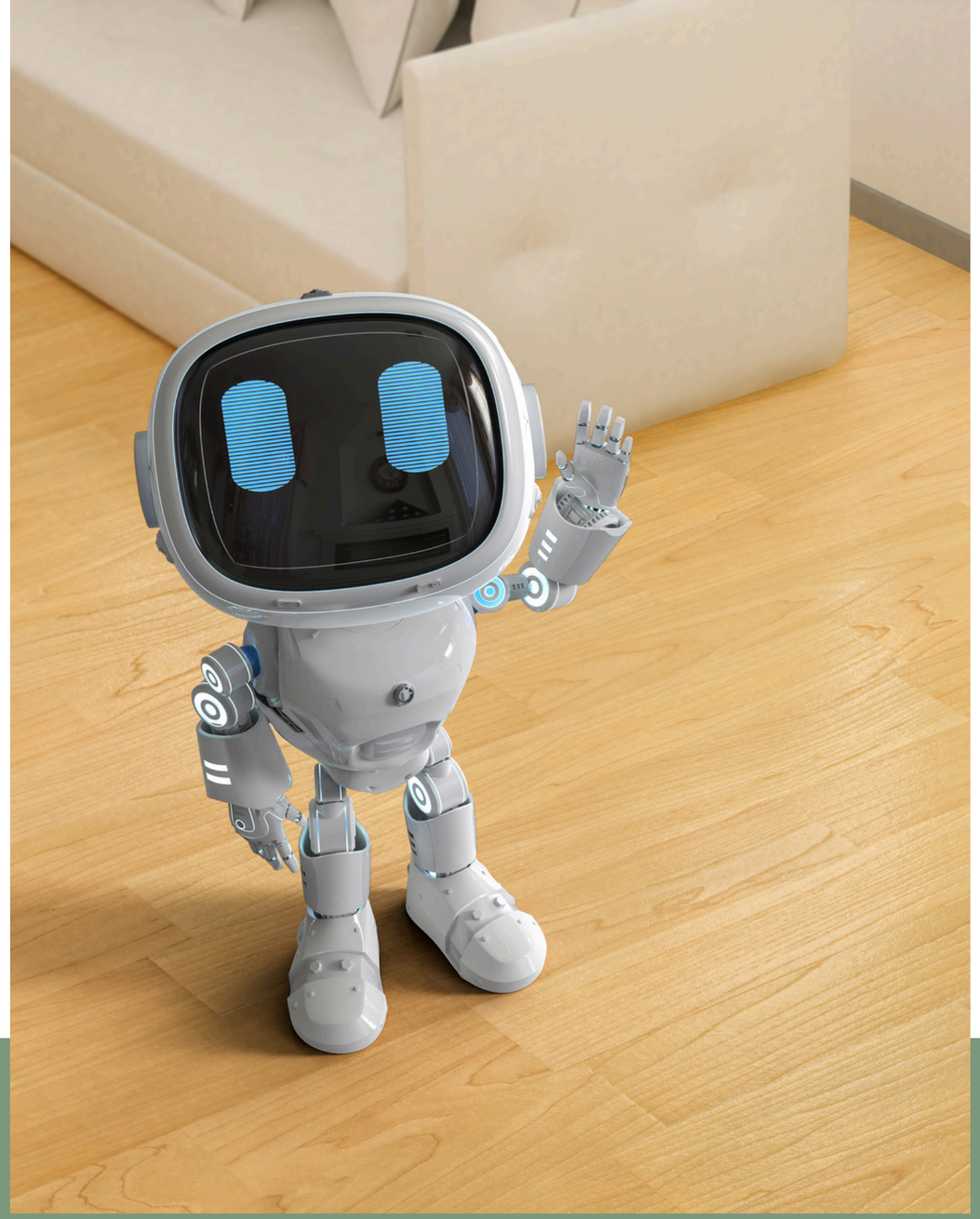

Introduction

- In today's world, food choices are abundant but often overwhelming. People struggle to decide what to cook based on what they have, how nutritious their meals are, or how to find similar alternatives to their favorite dishes.
- Our project, NomNom, is an intelligent culinary assistant that bridges this gap. By leveraging AI and intuitive interactions, it helps users discover recipes, analyze nutritional content, and even generate surprise dishes based on available ingredients.
- This bot not only personalizes food suggestions but also promotes health-conscious and engaging cooking experiences, all within a simple, chat-like interface.

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Background and Objectives

“NomNom aims to bridge culinary creativity with convenience.”

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Background

- With the growing demand for health-conscious eating, recipe discovery, and AI-powered personalization, the food tech industry is evolving rapidly.
- Users often rely on multiple apps—some for meal ideas, some for nutrition tracking, and others for voice assistants. But few platforms offer a unified and interactive solution tailored to the user’s pantry, dietary needs, and curiosity.
- Chatbots powered by natural language interaction have gained popularity due to their ease of use, making them ideal for a task like smart recipe generation and nutritional guidance.

Objectives

The primary goal of NomNom is to:

- Simplify meal decisions using intelligent recipe suggestions.
- Leverage AI to generate similar or surprise recipes based on user input.
- Analyze nutritional content of ingredients instantly.
- Provide a chat-based, voice-enabled experience that feels personal and intuitive.
- Present interactive elements like charts and step-by-step cooking instructions to enhance user engagement.

	Frontend Interface	Backend Processing (Flask)	NLP & Vector Search Pipeline
Methodology	• Built using HTML, CSS, JavaScript • Dynamic chat UI with support for buttons, speech synthesis, and user prompts • Chart.js for rendering interactive nutrition charts	• /lucky → AI-generated recipe using T5 model • /similar → Semantic search with: • SentenceTransformer embeddings • Cosine Similarity for matching input with recipe corpus • Indexed and retrieved using FAISS • /nutrition → Nutrition breakdown from Spoonacular API	• Input query is converted to an embedding via SentenceTransformer • Each recipe title & description is pre-embedded and stored • Cosine similarity scores are computed between the query and recipes • Top-N similar recipes retrieved using FAISS index for fast vector search

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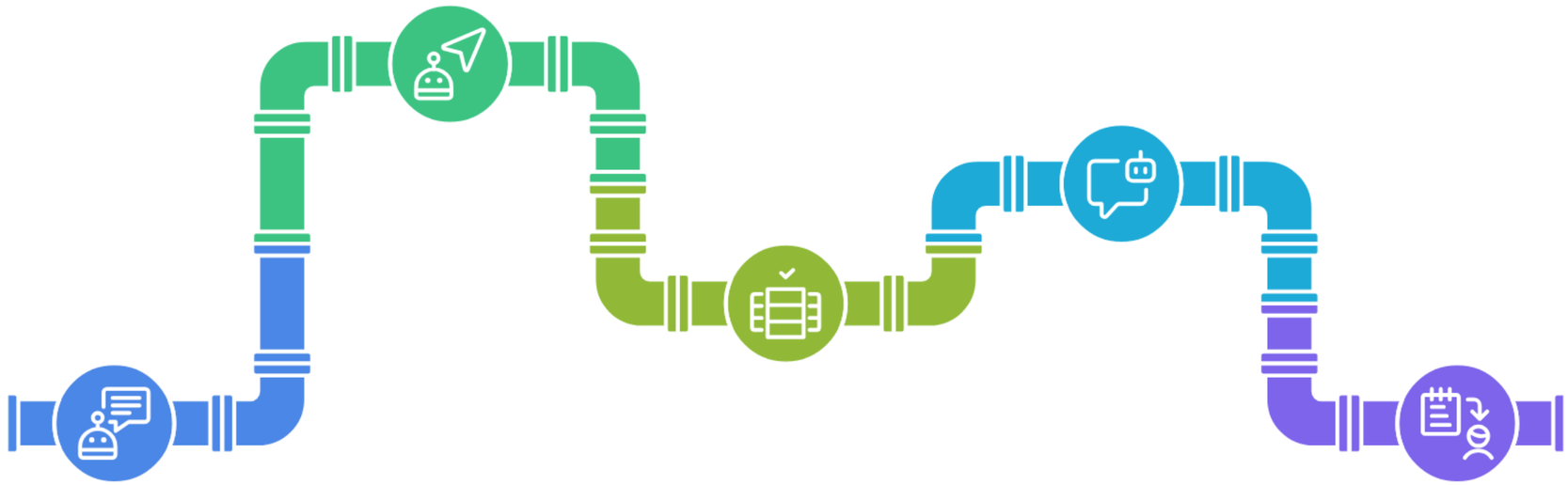
IMPLEMENTATION



01

User Input

User interacts with the chatbot via text or voice.

02

Bot Route Selection

Chatbot decides the route based on user intent. (Recipe Generation ,Recipe Similarity Search and Nutrition Info)

03

Backend Processing

Inputs are processed to generate recipes or analyze nutrition.

04

Chatbot Response Output

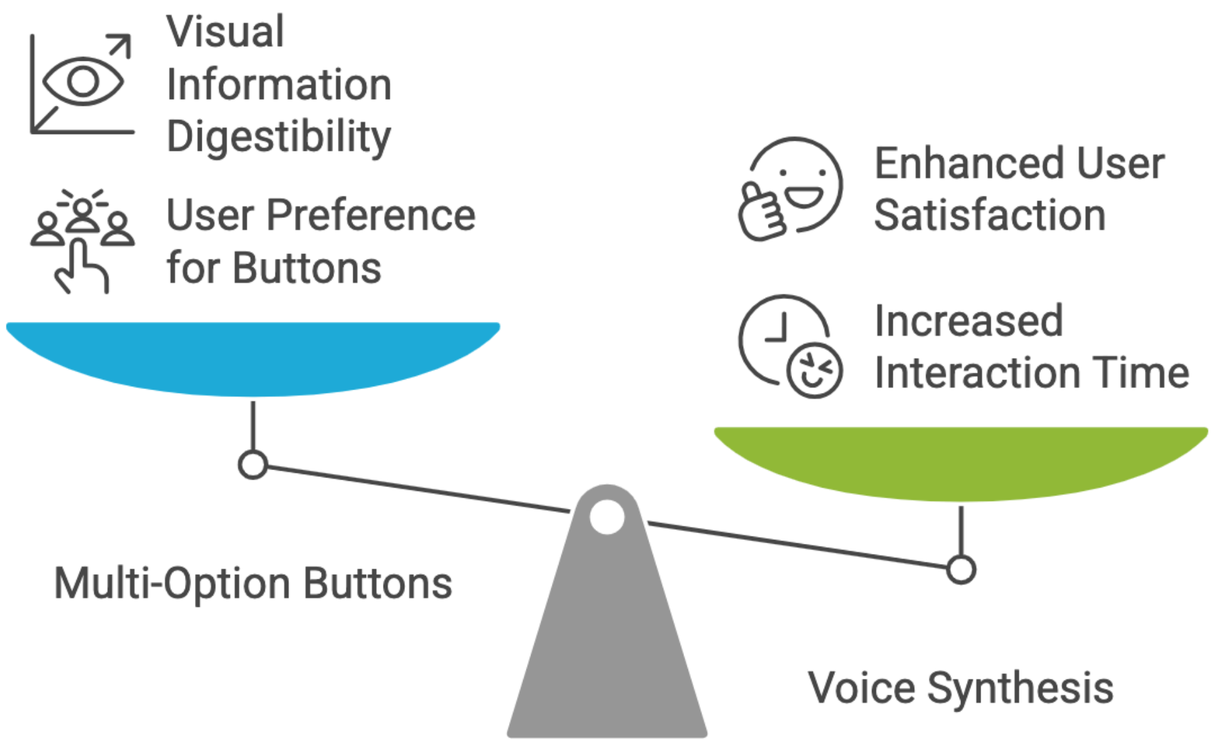
Chatbot responds with text, buttons, and visual aids.

05

Follow-up Context Handling

Chatbot maintains context for further interactions.

Analysis



Balancing User Interaction Preferences

Interactive Flow Boosts Engagement

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Analysis

T5 Recipe Generation Results

Choose the best method for recipe generation and presentation.

T5 Model

Generates coherent recipes but may lack precision

VS

Regex Parsing

Provides structured steps but requires manual input

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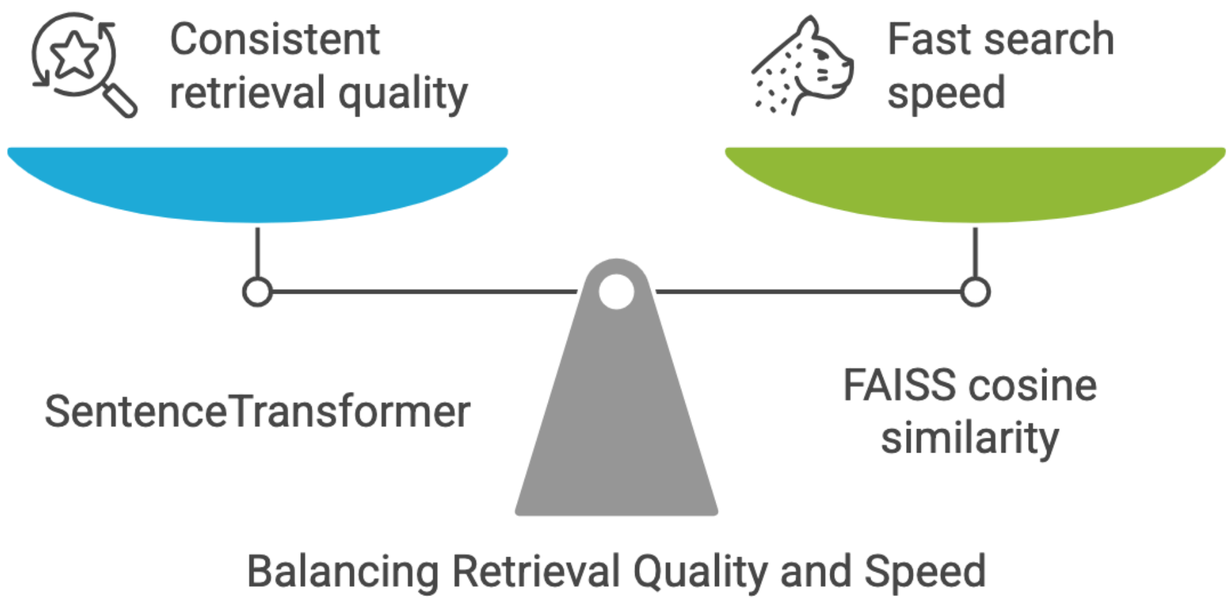
Analysis



Similarity Retrieval Accuracy

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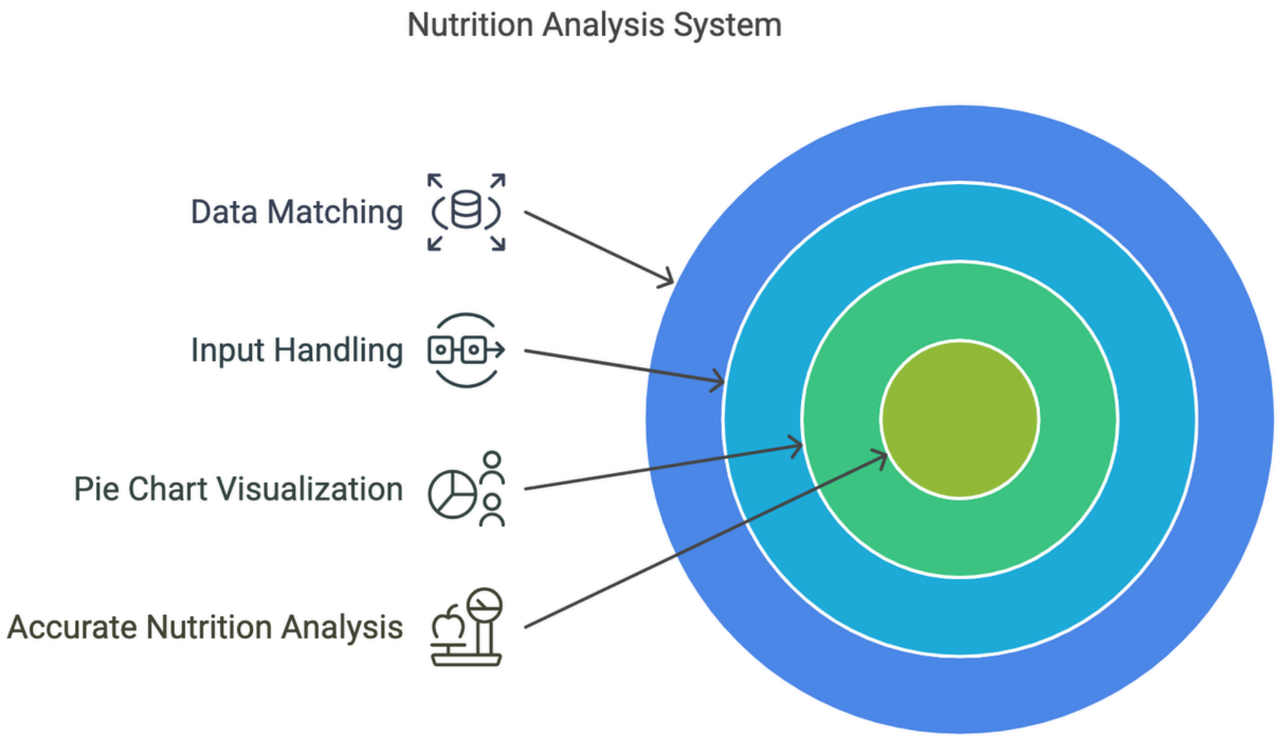
Analysis

Nutrition Information Precision



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Analysis

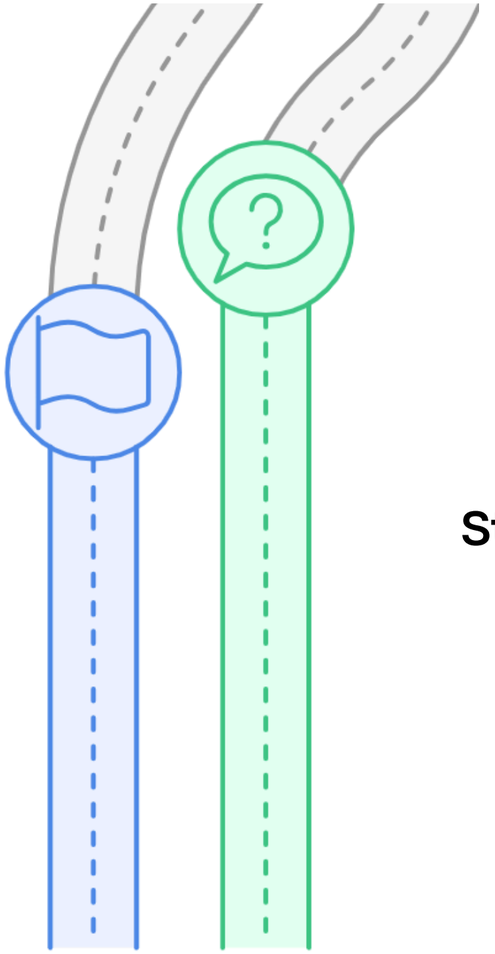
How to manage user interaction flow?

Use Flags

Ensures smooth transitions between modes, enhancing system reliability.

Dynamic Prompts

Improves user experience by adjusting prompts based on context.



State-Based Intent Handling

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Results

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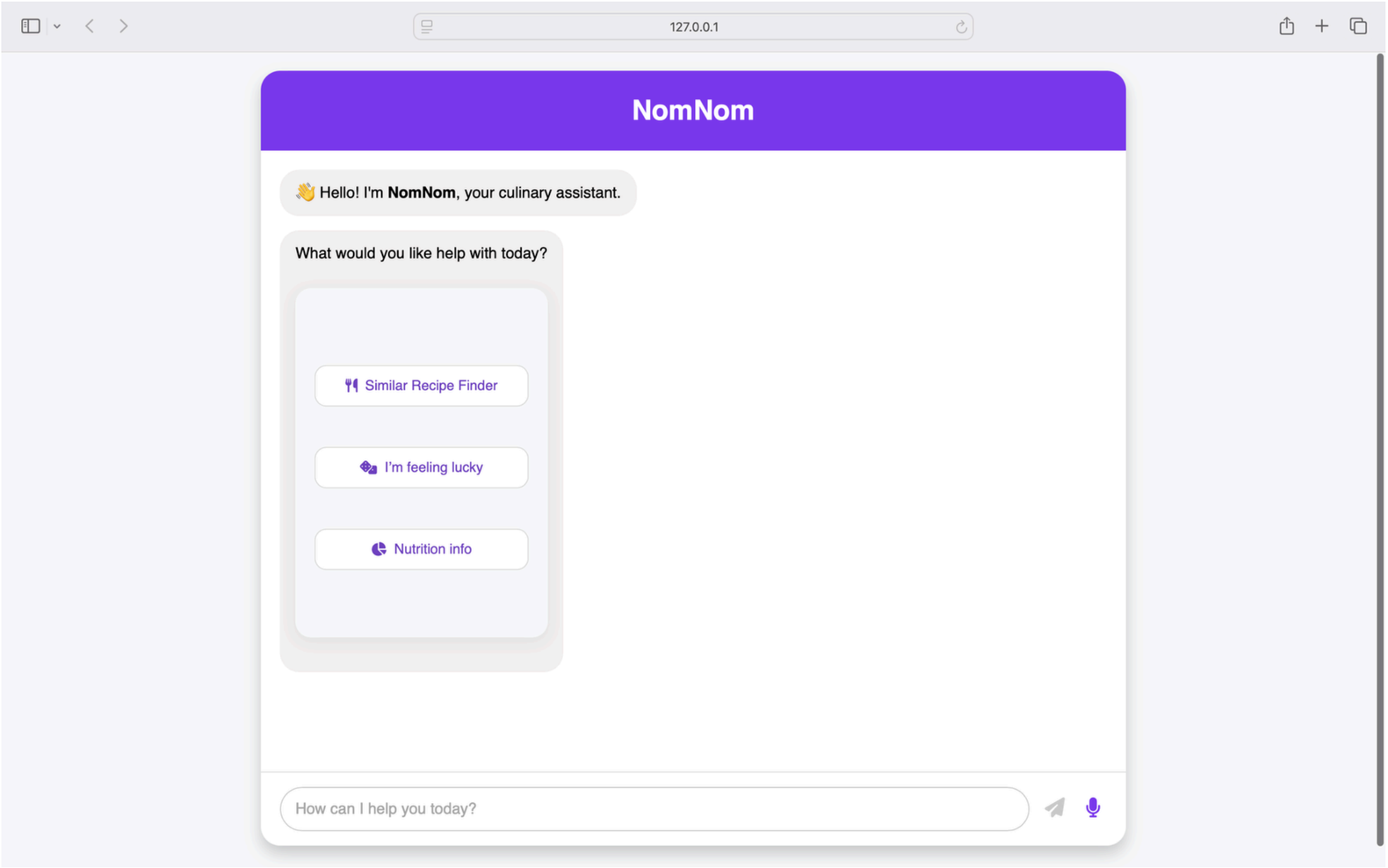

Landing page where the chatbot greets the user

Results

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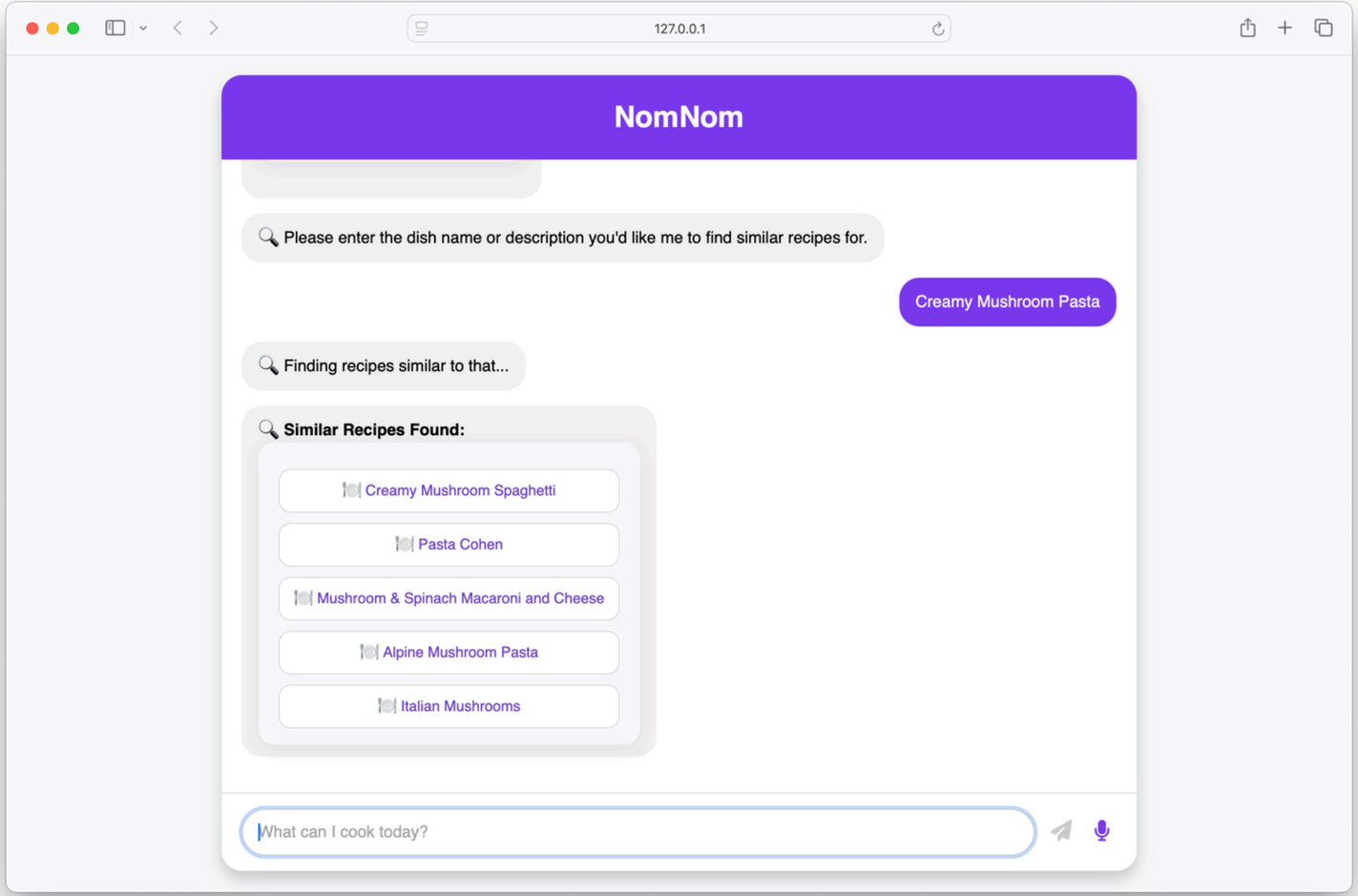

When clicked on “Find similar Recipes” , the module returns similar recipes based on user’s input

Results

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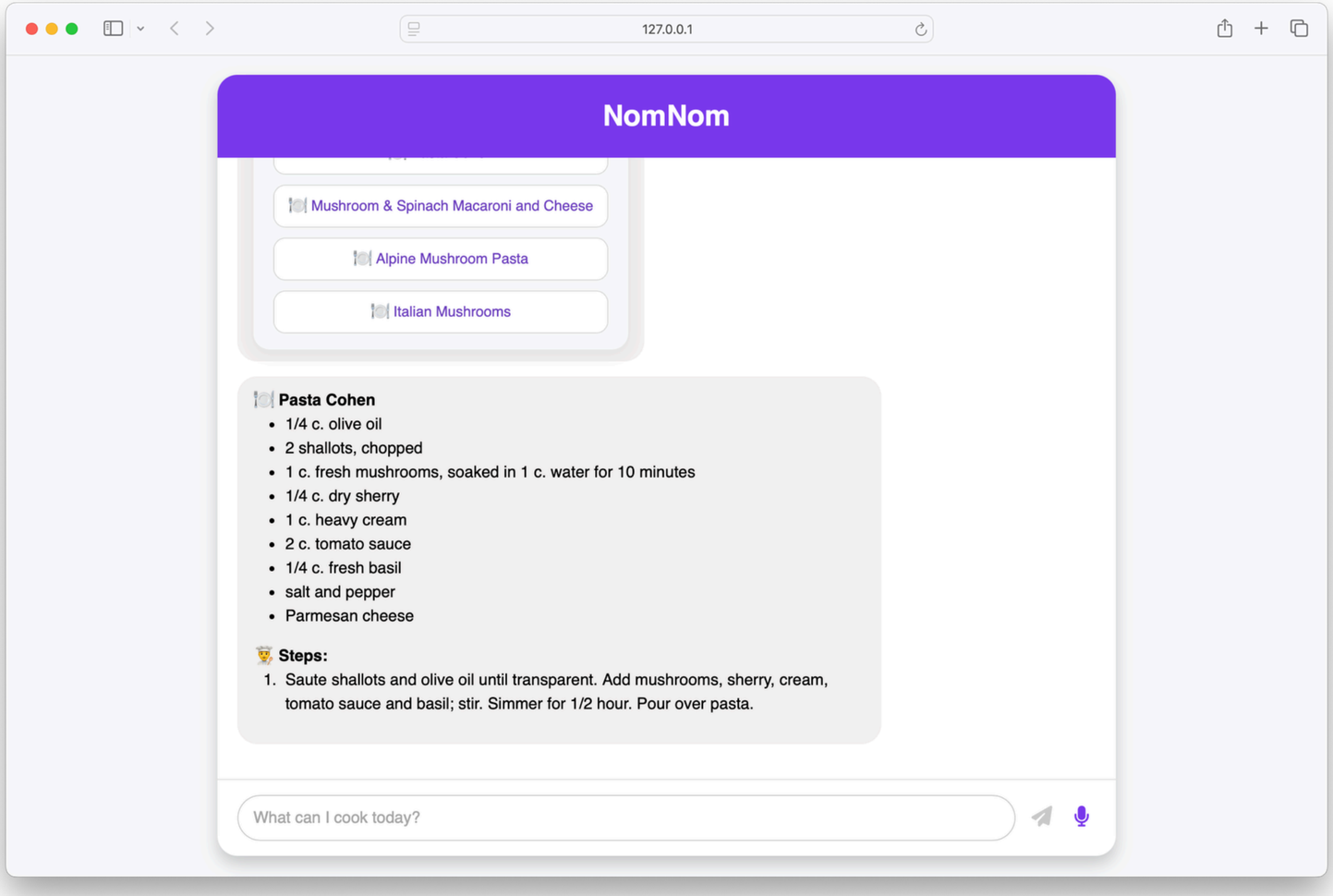

When clicked on the desired recipe , the ingredients and procedure is revealed

Results

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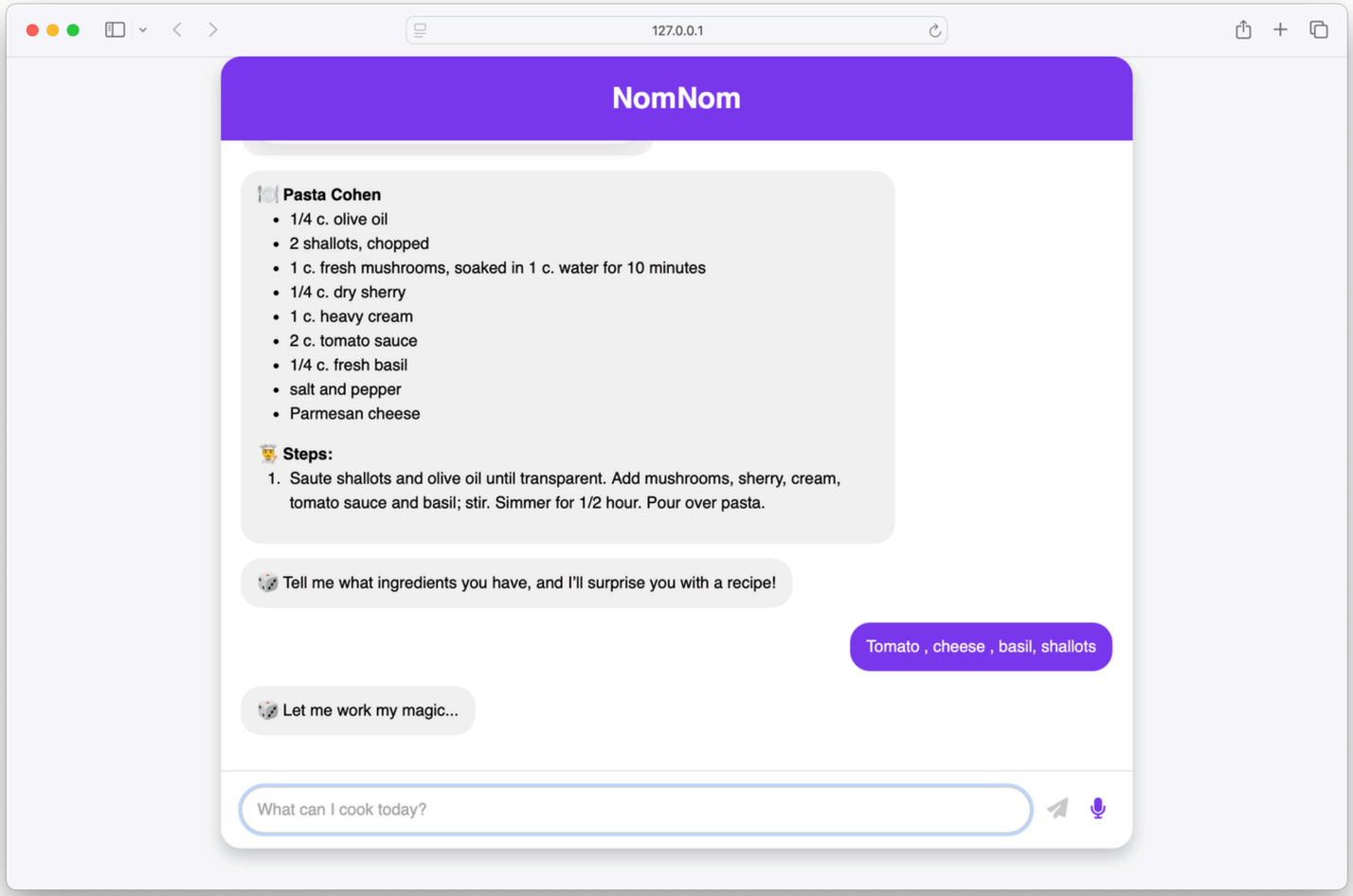

When the user clicks on “I’m feeling Lucky” component , based on the user’s input a random recipe is generated

Results

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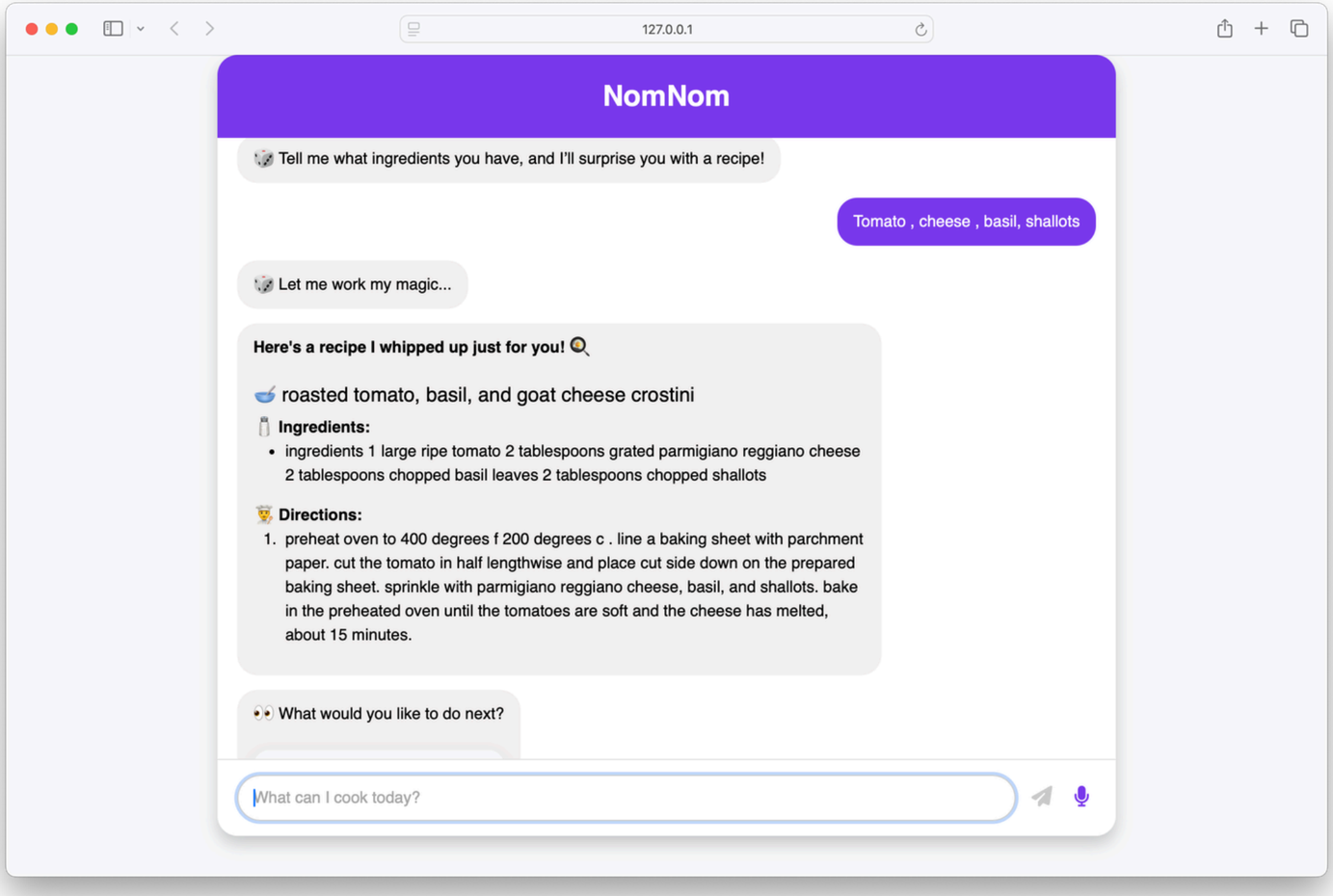

The recipe generated by the component that uses fine tuned t5 bert transformer on recipe dataset

Results

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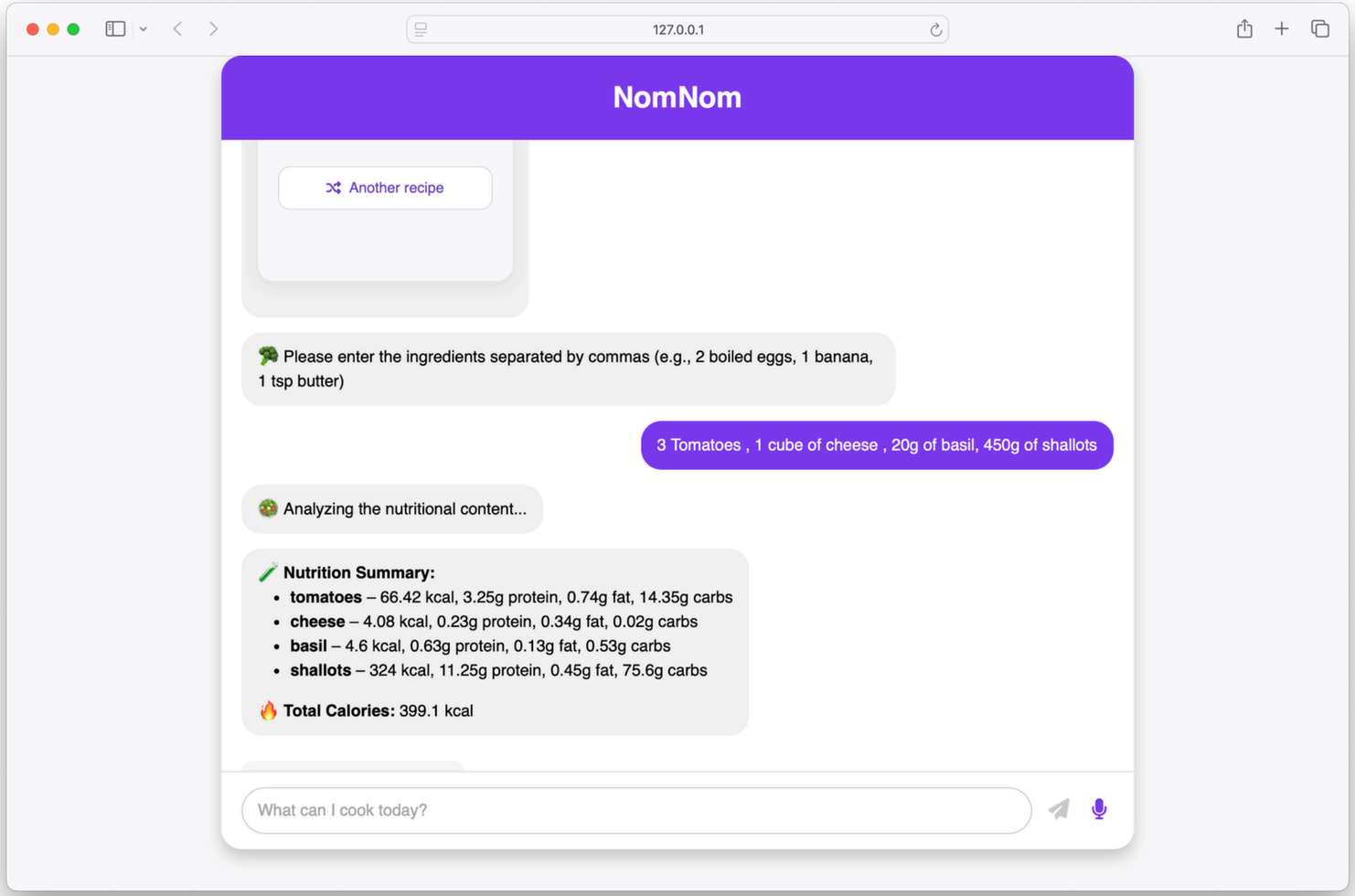

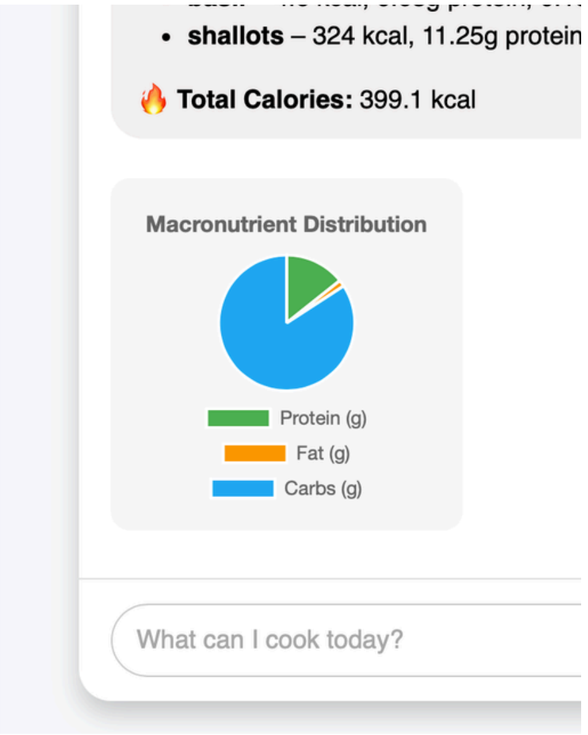

When the user wants the nutritional info , the ingredients along with their proportions are entered and summary is provided with a pie chart

Challenges

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01

Handling Multiple User Intents Simultaneously

Introduced state flags (awaitingNutrition, awaitingSimilarRecipe, etc.) to keep track of active intent and restrict actions until processed.

02

Overlapping or Incorrect Message Parsing

Used regex-based parsing and added fallback checks to extract and separate title, ingredients, and steps.

03

Speech-to-Text or TTS Mismatch

Added text feedback alongside bot speech and made speech optional to allow user correction.

04

Visual Overlap in Responses

Wrapped charts in a message-styled container, defined fixed height/width, and adjusted spacing using Flexbox for cleaner display.

05

Vector Search Accuracy with Sentence Embeddings

- Used SentenceTransformer embeddings (all-MiniLM-L6-v2) for recipe encoding.
- Applied cosine similarity as the distance metric.
- Indexed precomputed vectors in FAISS and returned top 5 matches for display.

06

Integrating T5 Recipe Generator

- Needed tokenizer alignment between T5Tokenizer and model.
- Required setting max_length, do_sample, and num_beams carefully to avoid empty or excessively long outputs.
- Had to manage GPU/CPU fallback and cold-start latency issues for live generation.

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Future Scope

Thank You

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